

Escuela de
Ingeniería y Arquitectura
Universidad Zaragoza

Universidad de Zaragoza

Facultad de Ingeniería

Artificial Intelligence - Lab Session 4: Explainability and Model Compression

Jorge Ruiz González (826685)
Hugo López Sola (875455)

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1 Task 1: Baseline CNN

We used the AugCXR dataset to classify chest X-rays as NORMAL or PNEUMONIA. There are 7,200 training images, 1,800 validation images and 1,000 test images, with equal numbers from each class. We used grayscale because colour is not needed for these images, resized them to 128×128, and normalized them using the training data. Horizontal flips were disabled.

We trained ResNet-18 from scratch for 10 epochs with AdamW, a batch size of 64 and an initial learning rate of 0.0003. The best validation result was at epoch 7, with about 95.1% accuracy. Training accuracy later reached 100% without improving validation accuracy, which suggests some overfitting.

Test accuracy was **97.8%**. Recall was 97.6% for NORMAL and 98.0% for PNEUMONIA. The confusion matrix below shows 22 mistakes in total. Accuracy is useful here because the classes are balanced, but checking both classes also shows whether one is being missed more often.

Table 1: Confusion matrix on the test set.

True class / predicted class	NORMAL	PNEUMONIA
NORMAL	488	12
PNEUMONIA	10	490

2 Task 2: Explainability

Grad-CAM usually highlighted broad areas inside the chest, although some heatmaps also reached the borders or corners. In some wrong predictions, the attention was near an image boundary or a letter marker. This suggests that the model sometimes uses irrelevant features. Other mistakes still focused on the chest, so the heatmaps alone do not explain every error. Some wrong predictions also had high confidence.

The earlier layer, `layer2`, gave smaller, scattered hotspots. `layer4` gave smoother and broader areas, but less detail: its feature maps are only 4×4 for our input size. Grad-CAM++ gave similar main hotspots to Grad-CAM, with small differences in their shape and intensity.

3 Task 3: Compression

Int8 quantization gave the best balance in our notebook. Accuracy only changed from 97.8% to 97.7%, while the saved model became about four times smaller (44.8 to 11.3 MB). CPU time for a batch of 32 images fell from about 575 to 378 ms, giving roughly a 1.5× speed-up.

Table 2: Test accuracy before and after fine-tuning the pruned models.

Pruned weights	30%	50%	70%	90%
Test accuracy before fine-tuning	96.5%	85.6%	50.0%	50.0%
After two fine-tuning epochs	97.4%	97.3%	97.3%	97.1%

Pruning without fine-tuning broke down at 70% and 90%, but two training epochs recovered most of the accuracy. It did not reduce file size or give a CPU speed-up because the zero weights were still stored and processed in dense tensors. For int8 calibration, one batch gave 94.9% validation accuracy and ten batches gave 95.1%. One was already close; we kept ten for our final int8 model.

4 Task 4: MANOLO workflow

4.1 Results and comparison

We registered the saved model and a dataset containing 2,000 training images and 1,000 test images. The workflow ran inference, 50% magnitude pruning with two fine-tuning epochs, and then 8-bit quantization with one training epoch. All three tasks finished successfully.

Table 3: Accuracy in the notebook and the MANOLO workflow.

Experiment	Notebook accuracy	MANOLO accuracy
Original model	97.8%	97.8%
50% pruning + fine-tuning	97.3%	96.8%
Quantization	97.7%	96.6%

The original model gave the same accuracy and per-class recall in both runs. Pruning kept most of the accuracy. In the notebook we quantized the original model, whereas MANOLO quantized the already pruned model. Its final accuracy was 1.2 percentage points below the baseline. MANOLO also used fewer training images for fine-tuning, so the compression experiments are not identical.

Quantization made the saved model about 3.9× smaller in MANOLO, similar to the notebook. Pruning alone left its size unchanged.

Grad-CAM was enabled in the workflow, but its heatmap report is missing from the saved files. We could not visually compare it with the notebook results.

4.2 Extra measurements

MANOLO also recorded energy use, estimated CO_2 emissions, hardware utilization, confidence and inference throughput.

Table 4: Resource measurements recorded by MANOLO.

Step	Duration (s)	Energy (Wh)	CO_2 (g)
Inference	10.6	0.36	0.063
Pruning	12.1	0.52	0.090
Quantization	11.6	0.46	0.080

These measurements include compression and fine-tuning, so they do not show the energy saved during future use. Average inference batch latency was about 4 ms on the MANOLO node. This cannot be directly compared with the notebook's CPU timing because the hardware and setup differ.

4.3 What MANOLO adds

The notebook was useful for trying things and inspecting images. MANOLO registered the data and models, ran the workflow and stored results in MLflow. This makes experiments easier to repeat and compare, and helps keep track of which model was used.

Quantization was the most useful compression method here. Pruning recovered good accuracy after fine-tuning, but did not make the dense model smaller or faster. We worked on one dataset and did not carry out the optional task.

4.4 MLflow screenshots

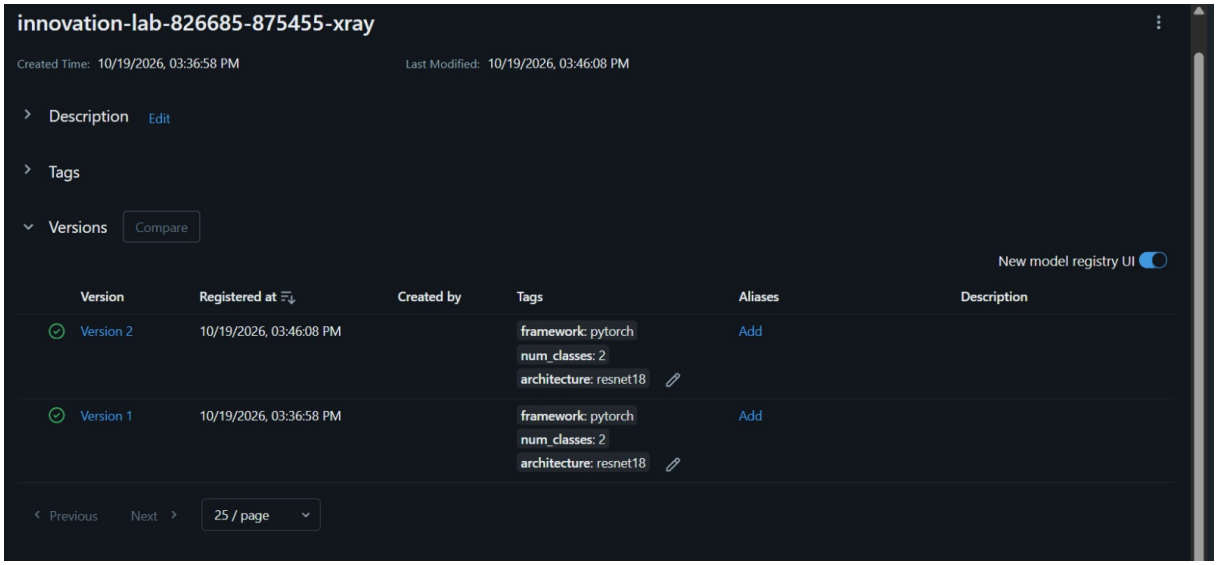


Figure 1: Model registered in MLflow. Version 2 was used in our workflow.

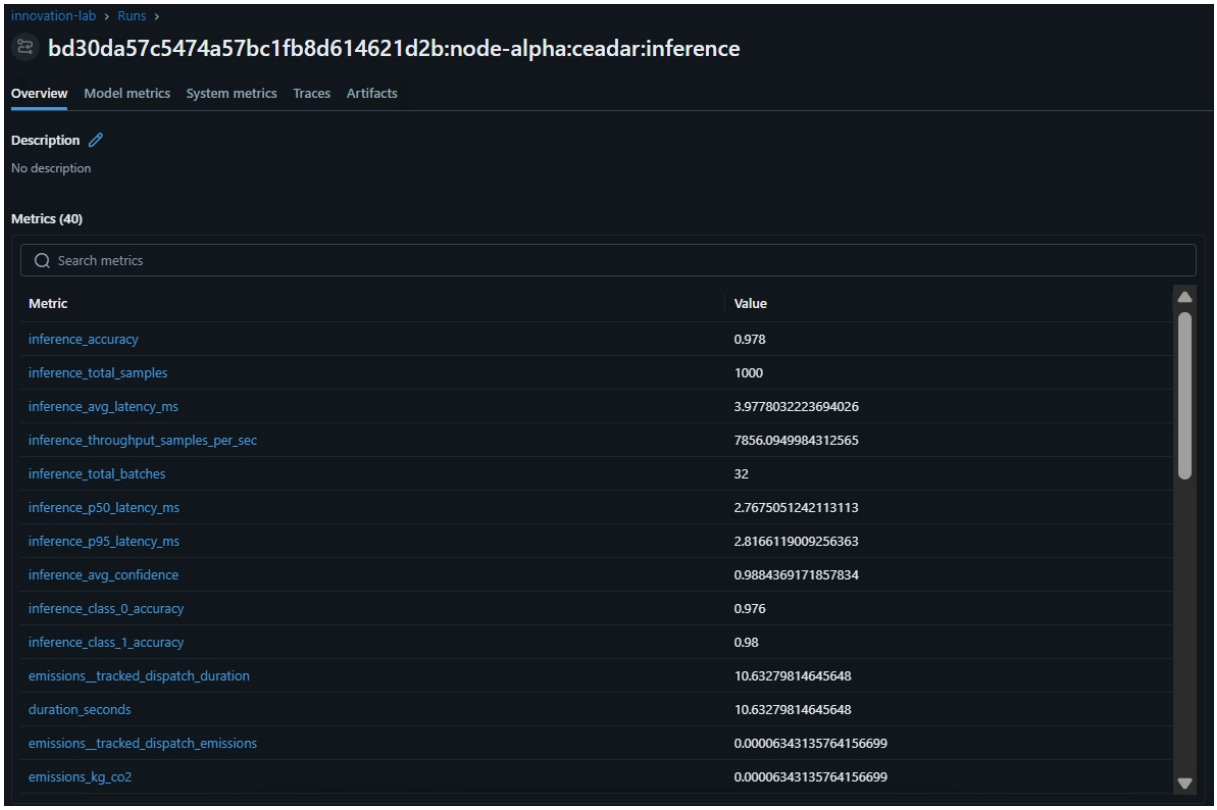


Figure 2: Inference results: 97.8% accuracy on the 1,000 test images.

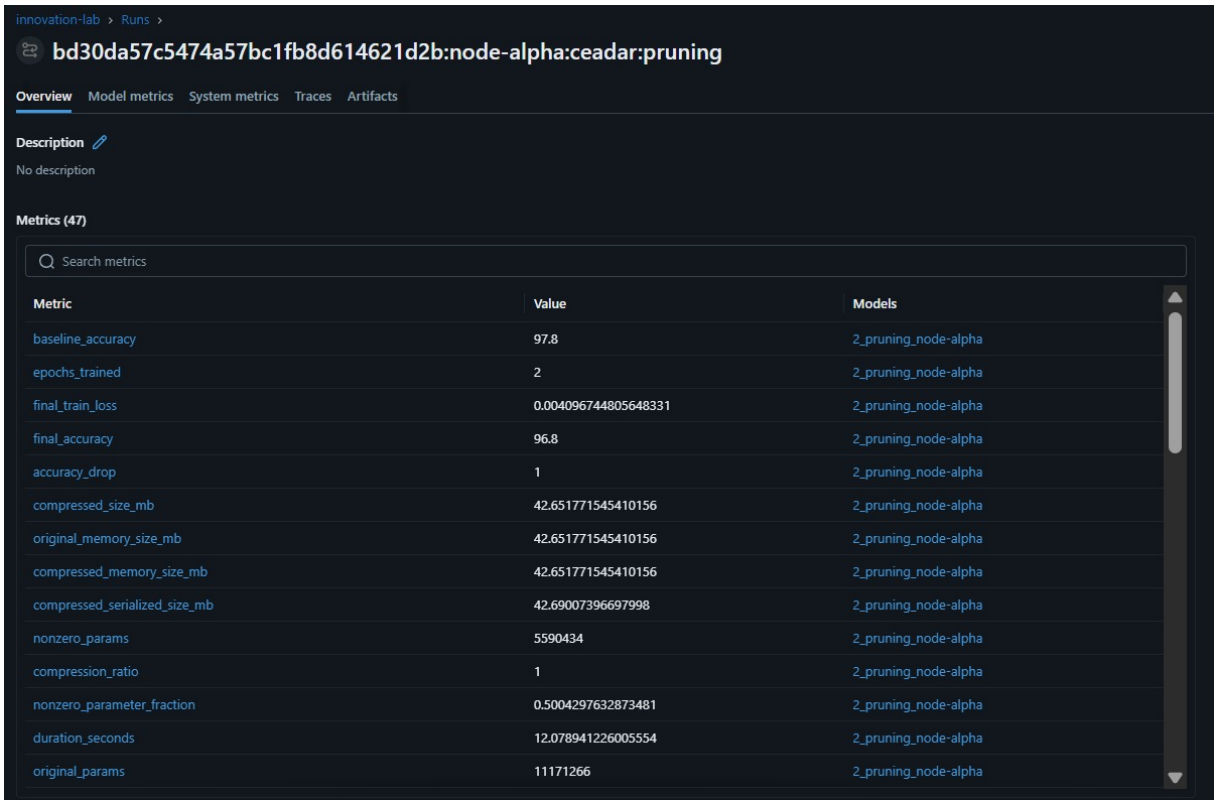


Figure 3: Pruning results: 96.8% accuracy after two fine-tuning epochs.

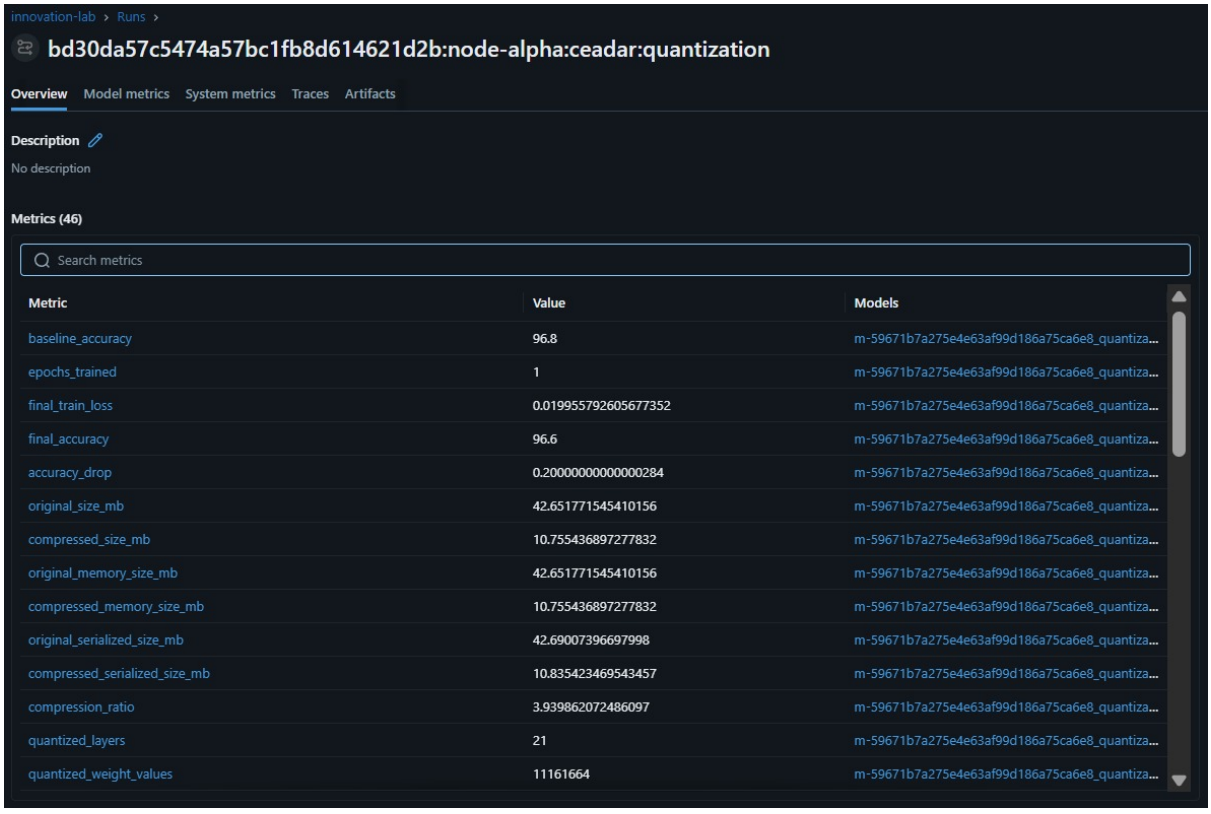


Figure 4: Quantization of the pruned model: 96.6% accuracy and about four times smaller size.